

Laptop Price Prediction Project Report

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[GitHub Repository](#)

Objective

The primary goal of this project was to develop a machine learning model that accurately predicts the price of laptops based on various features such as brand, RAM, storage, screen specifications, and more. This can assist consumers and retailers in understanding price trends and making informed purchase or listing decisions.

Dataset Information

- Source: Google Drive Dataset
- Total Records: 1,275
- Features: 23 columns
- Key Features Used:
 - Company
 - RAM
 - Operating System (OS)
 - Storage
 - Screen Size
 - IPS Panel
 - CPU
 - GPU
 - PPI (Pixels Per Inch)

Workflow & Methodology

1. Data Cleaning:

- Handled inconsistent formats and missing values.
- Reduced categorical granularity.

2. Exploratory Data Analysis (EDA):

- Visualized correlations between features and price.

3. Feature Engineering:

- Extracted PPI, simplified CPU/GPU into tiers.

4. Encoding & Normalization:

- Label Encoding & normalization techniques.

5. Model Training & Evaluation:

- Used Random Forest Regressor with hyperparameter tuning.

6. Deployment:

- Built Streamlit app and used Pickle for model persistence.

Model Performance

- Model: Random Forest Regressor
- R^2 Score: 0.8772
- MAE: 0.1584

Tools & Technologies Used

- Python, Pandas, NumPy, Scikit-learn
- Jupyter Notebook
- Streamlit

- Pickle

Challenges Faced

- CPU/GPU columns had many variants. Grouped into performance-based tiers.
- Extracted resolution & screen size from mixed-format column.

Conclusion

The model successfully predicts laptop prices with strong accuracy and generalization. This application can be used by consumers to estimate laptop prices or by businesses to benchmark and price products effectively.

Future Improvements

- Integrate XGBoost/LightGBM for comparison
- Add features like battery life, weight, and brand reputation
- Deploy with backend database for usage tracking